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Quantitative Sustainability: Conceptual frameworks

Learning outcomes of video

- Describe the conceptual causal relationship of environmental impacts
- Describe the importance of technology development in relation to other actions toward a more sustainable future

The sustainability challenge

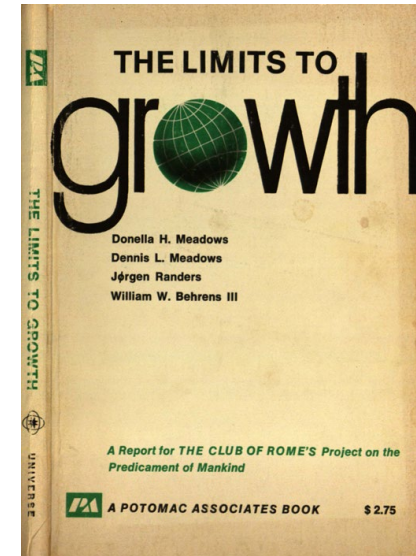
$$I = P \cdot A \cdot T = Pop \cdot \frac{GDP}{person} \cdot \frac{I}{GDP}$$

- I is the environmental impact
- Pop is the **global population**
- $\frac{GDP}{person}$ is the **Affluence**, the material standard of living
- $\frac{I}{GDP}$ is the **Technology factor** – environmental impact per created value

Ehrlich P, Holdren J (1971) Impact of population growth. Science 171, pp. 1212–1217.

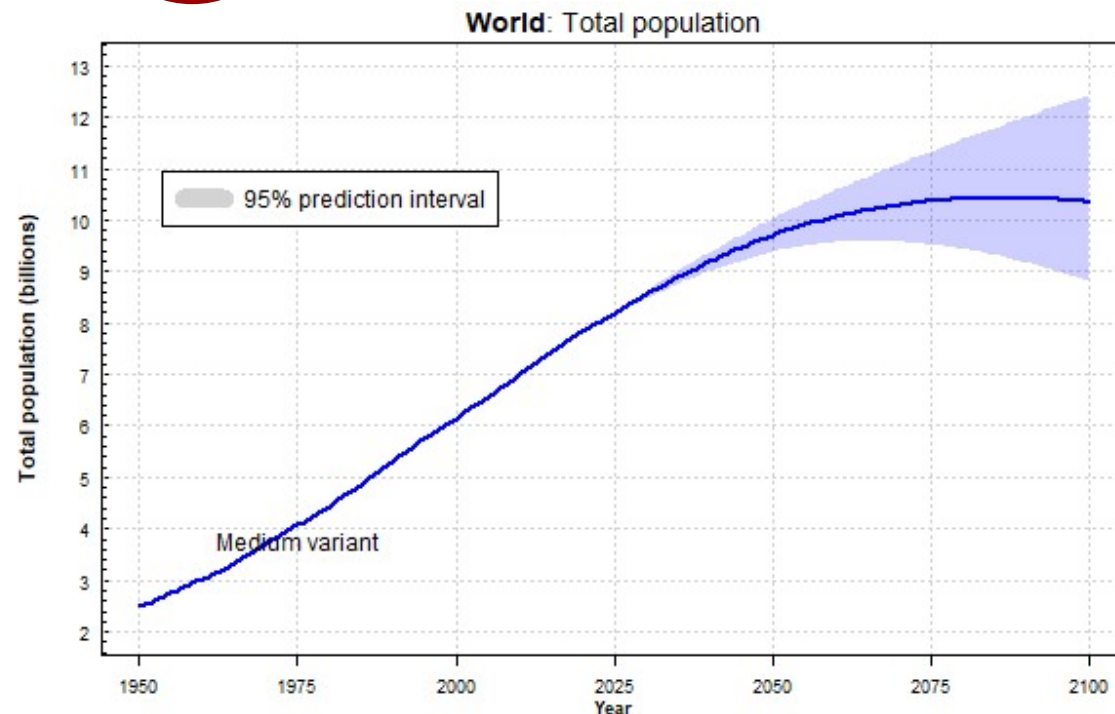
Commoner B (1972) The environmental cost of economic growth. In Ridker RG (ed.) Population, Resources and the Environment, pp. 339– 63. U.S. Government Printing Office, Washington, DC.

Graedel and Allenby (1995) Industrial ecology. Prentice Hall, New Jersey.



The global population is increasing

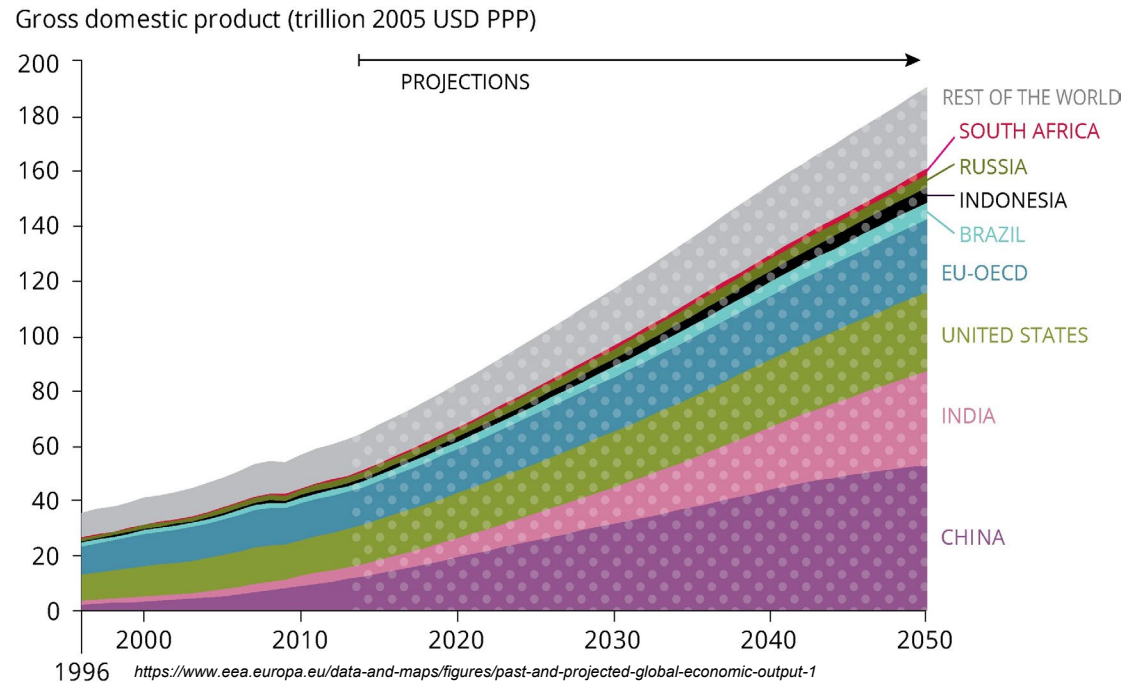
$$I = \text{Pop} \cdot \frac{GDP}{\text{person}} \cdot \frac{I}{GDP}$$



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United Nations, DESA, Population Division. *World Population Prospects 2022*. <http://population.un.org/wpp/>

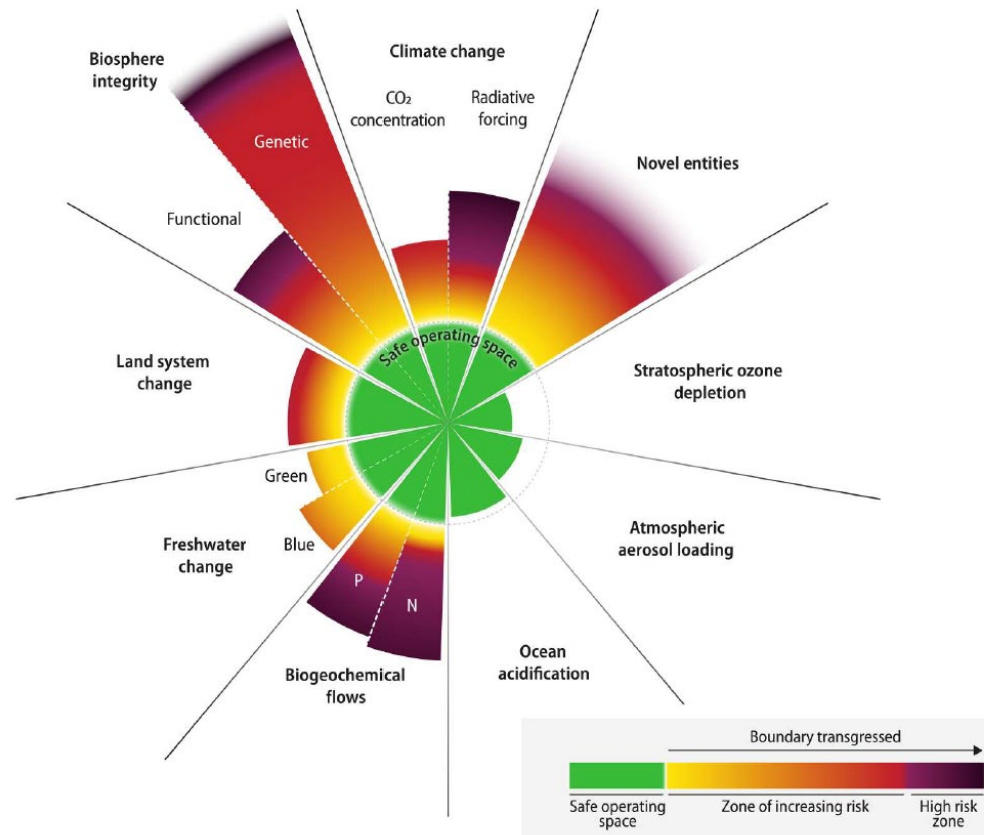
The affluence is increasing

$$I = Pop \cdot \frac{GDP}{person} \cdot \frac{I}{GDP}$$



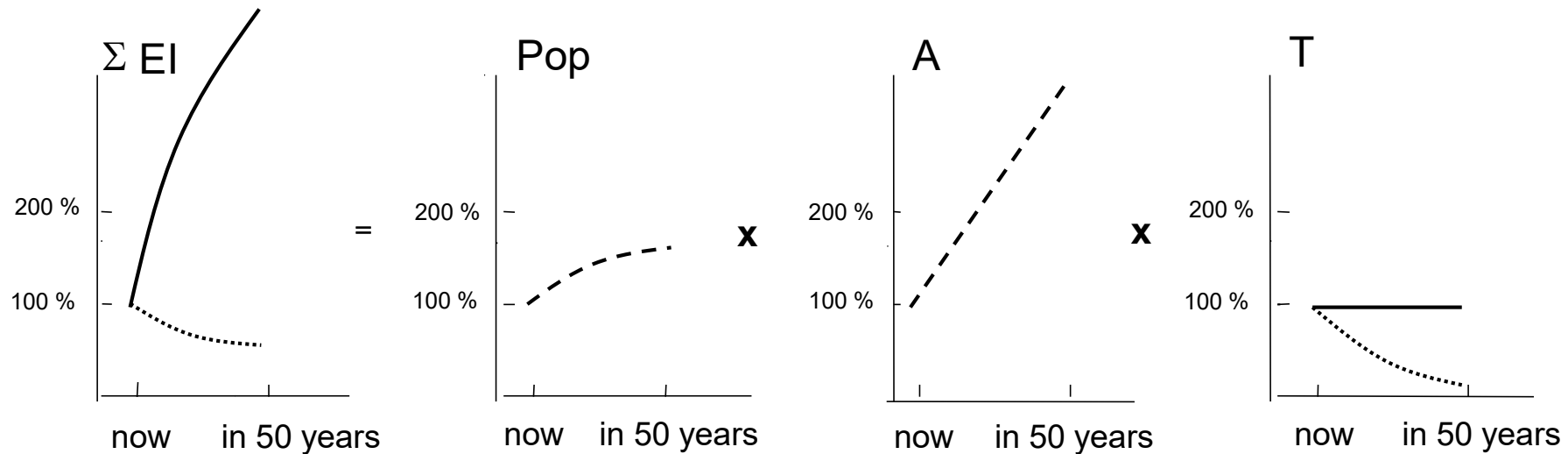
The environmental impacts should be reduced

$$I = \text{Pop} \cdot \frac{GDP}{\text{person}} \cdot \frac{I}{GDP}$$



Richardson et al., *Sci. Adv.* **9**, eadh2458 (2023)

How much should the technology factor be reduced?



$P \cdot A$ = the total human demand/consumption will grow by at least a factor of 4-5 in 50 years

ΣEI = the total human impact on the environment will grow accordingly, if the technology factor remains unchanged

$T \left(\frac{I}{GDP} \right)$ should be reduced by a factor 4 to 20 to achieve environmental sustainability

Need to address the consumption/demand

- Rebound effect
- Increasing human demands

Take aways

- The global environmental impact is driven by human consumption and demands
- We can achieve some reduction of the impact by technology development, which should be a factor 4-20 more efficient
- But we also need to address the human consumption by it self